

SINGLE CORRECT TYPE QUESTIONS

1. Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^n Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i 's and exactly 6 of sets Y_i 's, then n is equal to :

माना $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^n Y_i = T$ है, जहाँ प्रत्येक X_i में 10 अवयव हैं तथा प्रत्येक Y_i में 5 अवयव हैं। यदि T का प्रत्येक अवयव ठीक 20, X_i समुच्चयों का एक अवयव है तथा ठीक 6, Y_i समुच्चयों का एक अवयव है, तो n का मान है :

- (A) 30 (B) 50 (C) 15 (D) 45

Ans. A

Sol. Number of elements in $T = \frac{50 \times 10}{20} = \frac{n \times 5}{6}$
 $\Rightarrow n = 30$

2. Let $f(x) = \sin^{-1} x$ and $g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$. If $g(2) = \lim_{x \rightarrow 2} g(x)$, then the domain of the function fog is :

माना $f(x) = \sin^{-1} x$ तथा $g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$ हैं। यदि $g(2) = \lim_{x \rightarrow 2} g(x)$, तो फलन fog का प्रांत है :

- (A) $(-\infty, -1] \cup [2, \infty)$ (B) $(-\infty, -2] \cup [-1, \infty)$
 (C) $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$ (D) $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$

Ans. C

Sol. $g(2) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{2x^2 - x - 6}$
 $= \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(x-2)(2x+3)}$
 $= \lim_{x \rightarrow 2} \frac{(x+1)}{2x+3} = \frac{3}{7}$

$$g(x) = \begin{cases} \frac{x+1}{2x+3} & x \neq 2 \\ \frac{3}{7} & x = 2 \end{cases}$$

$$f \circ g(x) = \sin^{-1}(g(x))$$

for $f \circ g$ be defined

$$-1 \leq g(x) \leq 1$$

$$\therefore \text{ for } x = 2 \quad g(2) = \frac{3}{7}$$

$\therefore f \circ g$ is defined when $x = 2$ (i)

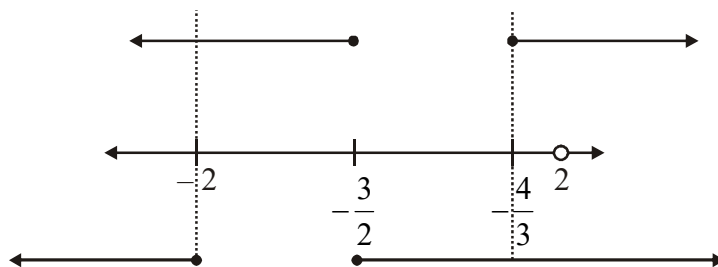
when $x \neq 2$

$$-1 \leq g(x) \leq 1$$

$$-1 \leq \frac{x+1}{2x+3} \leq 1$$

$$\frac{x+1}{2x+3} \geq -1 \text{ and } \frac{x+1}{2x+3} \leq 1$$

$$\Rightarrow \frac{3x+4}{2x+3} \geq 0 \text{ and } \frac{x+2}{2x+3} \geq 0$$



$$\therefore x \in (-\infty, -2] \cup \left[\frac{4}{3}, \infty \right) - \{2\} \quad \text{.....(ii)}$$

Using (i) and (ii)

Domain of $f \circ g$ is :

$$(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$$

3. The value of $\lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2}$, where r is non-zero real number and $[r]$ denotes the greatest integer less than or equal to r , is equal to:

$$\lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2}, \text{ जहाँ } r \text{ एक शून्येतर वास्तविक संख्या है तथा } [r] \text{ महत्तम पूर्णांक } \leq r \text{ है, का मान बराबर}$$

है :

- (A) $2r$ (B) 0 (C) r (D) $\frac{r}{2}$

Ans. D

Sol. By sandwich concept

$$x - 1 < [x] \leq x$$

$$1. r - 1 < [1 \cdot r] \leq 1 \cdot r$$

$$2. r - 1 < [2 \cdot r] \leq 2 \cdot r$$

$$n. r - 1 < [n \cdot r] \leq n \cdot r$$

By summation

$$r \frac{n(n+1)}{2} - n < [r] + [2r] + \dots + [nr] \leq r(1 + 2 + \dots + n)$$

$$\frac{r(n+1)}{2} - n < [r] + [2r] + \dots + [nr] \leq \frac{r n(n+1)}{2} \text{ division by } n^2$$

$$\lim_{n \rightarrow \infty} \frac{-n}{n^2} < \lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2} \leq \lim_{n \rightarrow \infty} \frac{r n(n+1)}{2n^2}$$

$$\frac{r}{2} < \frac{[r] + [2r] + \dots + [nr]}{n^2} \leq \frac{r}{2}$$

$$= \frac{r}{2}$$

4. If $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}$; $0 < x < 1$, then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is :

यदि $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}$; $0 < x < 1$, तो $\cos\left(\frac{\pi c}{a+b}\right)$ का मान है :

- (A) $\frac{1-y^2}{y\sqrt{y}}$ (B) $1-y^2$ (C) $\frac{1-y^2}{2y}$ (D) $\frac{1-y^2}{1+y^2}$

Ans. D

Sol. Let $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c} = K$.

$$\begin{aligned}\cos\left(\frac{\pi c}{a+b}\right) &= \cos\left(\frac{\pi \frac{\tan^{-1} y}{K}}{\frac{\sin^{-1} x}{K} + \frac{\cos^{-1} x}{K}}\right) \\ &= \cos\left(\frac{\pi \tan^{-1} y}{\frac{\pi}{2}}\right) \\ &= \cos(2 \tan^{-1} y) \\ &= \frac{1-y^2}{1+y^2}\end{aligned}$$

5. Let $A = \{2, 3, 4\}$ and $B = \{8, 9, 12\}$. Then the number of elements in the relation $R = \{(a_1, b_1), (a_2, b_2)\} \in (A \times B, A \times B) : a_1 \text{ divides } b_2 \text{ and } a_2 \text{ divides } b_1\}$ is :

माना $A = \{2, 3, 4\}$ तथा $B = \{8, 9, 12\}$ हैं। तो संबंध $R = \{(a_1, b_1), (a_2, b_2)\} \in (A \times B, A \times B) : a_1, b_2 \text{ को विभाजित करता है तथा } a_2, b_1 \text{ को विभाजित करता है}\}$ में अवयवों की संख्या है :

- (A) 24 (B) 12 (C) 18 (D) 36

Ans. D

Sol. $A = \{2, 3, 4\}, B = \{8, 9, 12\}$

$$a_1 \in A, b_2 \in B$$

$$a_1 \text{ divides } b_2$$

$$(a_1, b_2) \in \{(2, 4), (2, 12), (3, 9), (3, 12), (4, 8), (4, 12)\}$$

$$a_2 \in A, b_1 \in B$$

$$a_2 \text{ divides } b_1$$

.... same as above

$$\therefore \text{Number of relations} = 6 \times 6 = 36$$

6. The relation $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$ is :

(A) symmetric but not transitive

(B) transitive but not reflexive

(C) reflexive but not symmetric

(D) neither symmetric nor transitive

संबंध $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$:

(A) सममित है परन्तु संक्रामक नहीं है

(B) संक्रामक है परन्तु स्वतुल्य नहीं है

(C) स्वतुल्य है परन्तु सममित नहीं है

(D) न तो सममित है न ही संक्रामक है

Ans. D

Sol. (A) Reflexive

$$\text{let } a \in \mathbb{Z}$$

$$aRa \Rightarrow \gcd(a, a) = a$$

$\Rightarrow R$ is not reflexive

(B) Let $a, b \in \mathbb{Z}$ and aRb

$$\text{So, } aRb \Rightarrow \gcd(a, b) = 1$$

$$\Rightarrow \gcd(b, a) = 1, \forall a, b \in \mathbb{Z}$$

$$\Rightarrow bRa, \forall a, b \in \mathbb{Z} \Rightarrow R \text{ is symmetric relation on } \mathbb{Z}$$

7. Let $f : \mathbb{R} - \{0, 1\} \rightarrow \mathbb{R}$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$. Then $f(2)$ is equal to :

माना फलन $f : \mathbb{R} - \{0, 1\} \rightarrow \mathbb{R}$ इस प्रकार है कि $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$ है। तो $f(2)$ बराबर है :

(A) $\frac{9}{4}$

(B) $\frac{7}{4}$

(C) $\frac{9}{2}$

(D) $\frac{7}{3}$

Ans. A

Sol. , $x = 2 \Rightarrow f(2) + f(-1) = 3$... (1)

$$, x = \frac{1}{2} \Rightarrow f\left(\frac{1}{2}\right) + f(2) = \frac{3}{2} \quad \dots (2)$$

$$, x = -1 \Rightarrow f(-1) + f\left(\frac{1}{2}\right) = 0 \quad \dots(3)$$

$$(B) \dots\dots\dots (3) \Rightarrow f(2) - f(-1) = \frac{3}{2} \quad \dots(4)$$

$$(1) + (4) \Rightarrow 2f(2) = \frac{9}{2} \Rightarrow f(2) = \frac{9}{4}$$

8. Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n+1) = f(n) + f(1) \quad \forall n \in \mathbb{N}$ and g be any arbitrary function. Which of the following statements is NOT true ?

(A) If g is onto, then fg is one-one (B) If f is onto, then $f(n) = n \quad \forall n \in \mathbb{N}$

(C) If fg is one-one, then g is one-one (D) f is one-one

माना $f, g : \mathbb{N} \rightarrow \mathbb{N}$ हैं, जिनके लिए $f(n+1) = f(n) + f(1) \quad \forall n \in \mathbb{N}$ है तथा g एक स्वेच्छ फलन है ? निम्न में से कौन सा कथन सत्य नहीं है ?

(A) यदि g आच्छादक है, तो fg एकैकी है (B) यदि f आच्छादक है, तो $f(n) = n \quad \forall n \in \mathbb{N}$ है

(C) यदि fg एकैकी है, तो g एकैकी है (D) f एकैकी है

Ans. A

Sol. $f(n+1) = f(n) + f(1)$

$$n = 1$$

$$f(2) = f(1) + f(1)$$

$$f(2) = 2f(1)$$

$$f(3) = 3f(1)$$

$$f(4) = 4 + (1)$$

Option (1) : If 'g' is onto then fg might not be one - one, So it is wrong

Option (2) : $f(n) = n$

$$f(1) = 1$$

$$f(2) = 2$$

$$f(3) = 3$$

hence onto which is correct.

Option (3) : Correct, because fog is one - one them 'g' should by one - one.

Option (4) : f is one - one, hence this is also correct.

9. Let N be the set of natural numbers and two functions f and g be defined as $f, g : N \rightarrow N$ such that

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases} \text{ and } g(n) = n - (-1)^n. \text{ Then } fog \text{ is:}$$

(A) both one-one and onto

(B) one-one but onto

(C) onto but not one-one

(D) neither one-one nor onto

माना समुच्चय N प्राकृत संख्याओं को दर्शाता है तथा दो फलन f और g निम्न तरीके से परिभाषित है:

$$f, g : N \rightarrow N$$

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$$

तथा $g(n) = n - (-1)^n$; तो फलन fog

(A) एकैकी तथा आच्छादी दोनों हैं।

(B) एकैकी है परन्तु एकैकी नहीं है।

(C) आच्छादी है परन्तु एकैकी नहीं है।

(D) न आच्छादी है और न ही एकैकी है।

Ans. C

10. Let $A = \{1, 2, 3, \dots, 10\}$ and $f : A \rightarrow A$ be defined as

$$f(k) = \begin{cases} k+1 & \text{if } k \text{ is odd} \\ k & \text{if } k \text{ is even} \end{cases}$$

Then the number of possible functions $g : A \rightarrow A$ such that $gof = f$ is :

माना $A = \{1, 2, 3, \dots, 10\}$ है तथा $f : A \rightarrow A$,

$$f(k) = \begin{cases} k+1 & \text{यदि } k \text{ विषम है} \\ k & \text{यदि } k \text{ सम है} \end{cases}$$

द्वारा परिभाषित है। तो ऐसे फलनों $g : A \rightarrow A$, जिनके लिए $gof = f$ है, की सम्भावित संख्या है :

(A) $^{10}C_5$

(B) 10^5

(C) $5!$

(D) 5^5

Ans. B

Sol. $g(f(x)) = f(x)$

$\Rightarrow g(x) = x$, when x is even

so total number of functions g from A to A

$$= 10^5 \cdot 1$$

$$= 10^5$$

11. Let $S_k = \sum_{r=1}^k \tan^{-1} \left(\frac{6^r}{2^{2r+1} + 3^{2r+1}} \right)$. Then $\lim_{k \rightarrow \infty} S_k$ is equal to :

माना $S_k = \sum_{r=1}^k \tan^{-1} \left(\frac{6^r}{2^{2r+1} + 3^{2r+1}} \right)$ है, तो $\lim_{k \rightarrow \infty} S_k$ बराबर है :

(A) $\cot^{-1} \left(\frac{3}{2} \right)$ (B) $\tan^{-1} \left(\frac{3}{2} \right)$ (C) $\tan^{-1} (3)$ (D) $\frac{\pi}{2}$

Ans. A

Sol. $T_r = \tan^{-1} \left(\frac{\frac{6^r}{2^{2r+1}}}{1 + \frac{3^{2r+1}}{2^{2r+1}}} \right)$

$$= \tan^{-1} \left(\frac{\left(\frac{3}{2} \right)^{r+1} - \left(\frac{3}{2} \right)^r}{1 + \left(\frac{3}{2} \right)^{r+1} \left(\frac{3}{2} \right)^r} \right)$$

$$T_r = \tan^{-1} \left(\frac{3}{2} \right)^{r+1} - \tan^{-1} \left(\frac{3}{2} \right)^r$$

$$S_k = \tan^{-1} \left(\frac{3}{2} \right)^{k+1} - \tan^{-1} \left(\frac{3}{2} \right)$$

$$k \rightarrow \infty$$

$$\lim_{k \rightarrow \infty} S_k = \frac{\pi}{2} - \tan^{-1} \left(\frac{3}{2} \right) = \cot^{-1} \left(\frac{3}{2} \right)$$

12. Let $[t]$ denote the greatest integer $\leq t$. If for some $\lambda \in \mathbf{R} - \{0, 1\}$, $\lim_{x \rightarrow 0} \left| \frac{1-x+|x|}{\lambda-x+[x]} \right| = L$, then L is equal to:

माना $[t]$ महत्तम पूर्णांक $\leq t$ को दर्शाता है। यदि $\lambda \in \mathbf{R} - \{0, 1\}$, $\lim_{x \rightarrow 0} \left| \frac{1-x+|x|}{\lambda-x+[x]} \right| = L$ है, तो L का मान है :

- (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Ans. B

Sol. $\lim_{x \rightarrow 0} \left| \frac{1-x+|x|}{\lambda-x+[x]} \right| = L$

$$\text{LHL } \lim_{x \rightarrow 0^-} \left| \frac{1-x-x}{\lambda-x-1} \right|$$

$$\Rightarrow \frac{1}{|\lambda-1|}$$

$$\text{RHL } \lim_{x \rightarrow 0^+} \left| \frac{1-x+x}{\lambda-x+0} \right|$$

$$\Rightarrow \frac{1}{|\lambda|}$$

$$\text{LHL} = \text{RHL} = L$$

$$|\lambda| = |\lambda-1|$$

$$\lambda = \frac{1}{2}$$

$$L = \frac{1}{\left(\frac{1}{2}\right)} = 2$$

13. The value of $\cot \left(\sum_{n=1}^{19} \cot^{-1} \left(1 + \sum_{p=1}^n 2p \right) \right)$ is :

$$\cot \left(\sum_{n=1}^{19} \cot^{-1} \left(1 + \sum_{p=1}^n 2p \right) \right) \text{ का मान है}$$

(A) $\frac{22}{23}$

(B) $\frac{19}{21}$

(C) $\frac{23}{22}$

(D) $\frac{21}{19}$

Ans. D

Sol. $\sum_{n=1}^{19} \cot^{-1} \left(1 + 2 \cdot \frac{n(n+1)}{2} \right)$

$$\sum_{n=1}^{19} \cot^{-1} (1 + n(n+1))$$

$$\sum_{n=1}^{19} \tan^{-1} \left(\frac{1}{1 + (n+1)n} \right) = \tan^{-1} \left[\frac{(n+1) - n}{1 + n(n+1)} \right]$$

$$\sum_{n=1}^{19} \tan^{-1} (\tan^{-1}(n+1) - \tan^{-1}(n))$$

$$\tan^{-1}(2) - \tan^{-1}(1)$$

$$\tan^{-1}(3) - \tan^{-1}(2)$$

$$\tan^{-1}(20) - \tan^{-1}(19)$$

$$\tan^{-1}(20) - \tan^{-1}(1)$$

$$\tan^{-1} \left(\frac{20-1}{1+20} \right) = \tan^{-1} \left(\frac{19}{21} \right) = \cot^{-1} \left(\frac{21}{19} \right)$$

$$\cot \cot^{-1} \left(\frac{21}{19} \right) = \frac{21}{19}$$

14. Let $x = 2$ be a root of the equation $x^2 + px + q = 0$ and

$$f(x) = \begin{cases} \frac{1 - \cos(x^2 - 4px + q^2 + 8q + 16)}{(x - 2p)^4}, & x \neq 2p \\ 0, & x = 2p \end{cases}$$

Then $\lim_{x \rightarrow 2p^+} [f(x)]$,

where $[\cdot]$ denotes greatest integer function, is

माना $x = 2$ समीकरण $x^2 + px + q = 0$ का एक मूल है और

$$f(x) = \begin{cases} \frac{1 - \cos(x^2 - 4px + q^2 + 8q + 16)}{(x - 2p)^4} & , x \neq 2p \\ 0 & , x = 2p \end{cases}$$

है। तब $\lim_{x \rightarrow 2p^+} [f(x)]$, जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है का मान है :

- (A) -1 (B) 0 (C) 2 (D) 1

Ans. B

Sol. $\lim_{x \rightarrow 2p^+} \left(\frac{1 - \cos(x^2 - 4px + q^2 + 8q + 16)}{(x^2 - 4px + q^2 + 8q + 16)^2} \right) \left(\frac{(x^2 - 4px + q^2 + 8q + 16)^2}{(x - 2p)^2} \right)$

$$\lim_{h \rightarrow 0} \frac{1}{2} \left(\frac{(2p+h)^2 - 4p(2p+h) + q^2 + 8q + 16}{h^2} \right)^2 = \frac{1}{2}$$

Using L'Hospital's

$$\lim_{x \rightarrow 2p^+} [f(x)] = 0$$

15. Let $f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$

Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S.

- (A) equals $\{-2, -1, 1, 2\}$ (B) equals $\{-2, 2\}$
(C) equals $\{-2, -1, 0, 1, 2\}$ (D) is an empty set

$$\text{माना } f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$$

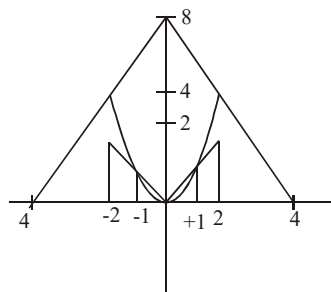
माना S, अन्तराल $(-4, 4)$ के उन बिन्दुओं, जिन पर f अवकलनीय नहीं है, का समुच्चय है, तो S :

- (A) $\{-2, -1, 1, 2\}$ के बराबर है। (B) $\{-2, 2\}$ के बराबर है।
(C) $\{-2, -1, 0, 1, 2\}$ के बराबर है। (D) एक रिक्त समुच्चय है।

Ans. C

Sol. $f(x) = \begin{cases} \max(|x|, x^2) & |x| \geq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$

$$\begin{cases} \max(|x|, x^2) & -2 \leq x \leq 2 \\ 8 - 2|x|, & x \in [-4, -2) \cup (2, 4] \end{cases}$$



ND points $x = -2, -1, 0, 1, 2$

16. The value of $\lim_{n \rightarrow \infty} \frac{1+2-3+4+5-6+\dots+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is :

$\lim_{n \rightarrow \infty} \frac{1+2-3+4+5-6+\dots+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ का मान है :

- (A) $3(\sqrt{2}+1)$ (B) $\frac{3}{2}(\sqrt{2}+1)$ (C) $\frac{\sqrt{2}+1}{2}$ (D) $\frac{3}{2\sqrt{2}}$

Ans. B

Sol.
$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n ((3r-2) + (3r-1) - 3r)}{\sqrt{2n^4+3n+1}-\sqrt{n^4+n+3}}$$

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n 3(r-1)}{\sqrt{2n^4+3n-1}-\sqrt{n^4+n+3}}$$

$$= \lim_{n \rightarrow \infty} \frac{3 \frac{n(n-1)}{2} (\sqrt{2n^4+3n-1} + \sqrt{n^4+n+3})}{(2n^4+3n-1) - (n^4+n+3)}$$

$$= \frac{3}{2}(\sqrt{2}+1)$$

17. $\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ is equal to :

$\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ बराबर है :

- (A) $\frac{(n+1)}{2}$ (B) n^2 (C) $n^2 + n$ (D) n

Ans. D

Sol. $\lim_{t \rightarrow 0} \left(\frac{1}{n^{\frac{1}{\sin^2 t}}} \right)^{\sin^2 t} \left[1 + \left(\frac{1}{n} \right)^{\csc^2 t} + \left(\frac{2}{n} \right)^{\csc^2 t} + \dots + \left(\frac{n-1}{n} \right)^{\csc^2 t} \right]^{\sin^2 t}$

as $0 < \frac{1}{n} < 1$, So $\left(\frac{1}{n} \right)^{\infty} = 0$

So, $n [1 + 0]^0 = n$

18. Let $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$, $x \in \mathbb{R}$. If $f(x)$ attains maximum value at α and $g(x)$ attains minimum value at β then is

$\lim_{x \rightarrow \alpha\beta} \frac{(x-1)(x^2-5x+6)}{x^2-6x+8}$ equal to

माना $f(x) = 5 - |x - 2|$ तथा $g(x) = |x + 1|$, $x \in \mathbb{R}$. यदि $f(x)$ का अधिकतम मान α पर है तथा $g(x)$ का न्यूनतम मान β पर

है, तो $\lim_{x \rightarrow \alpha\beta} \frac{(x-1)(x^2-5x+6)}{x^2-6x+8}$ बराबर है :

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

Ans. A

sol. $f(x) = 5 - |x - 2|$

Graph of $y = f(x)$

$f(x)$ is maximum at $x = 2$

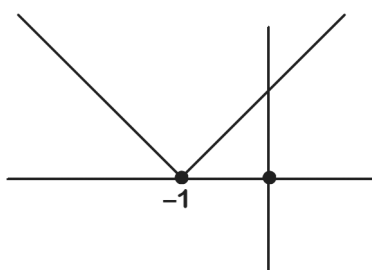
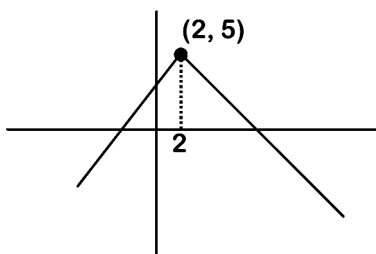
$\alpha = 2$

$g(x) = |x + 1|$

Graph of $y = g(x)$

$g(x)$ is minimum at $x = -1$

$\beta = -1$



$$\lim_{x \rightarrow 2} \frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)} = \lim_{x \rightarrow 2} \frac{(x-1)(x-3)}{x-4}$$

$$= \frac{1}{2}$$

19. Let $S = \{x : x \in \mathbb{R} \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10\}$. Then $n(S)$ is equal to :

माना $S = \{x : x \in \mathbb{R} \text{ एवं } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10\}$ है। तब $n(S)$ बराबर है :

- (A) 0 (B) 2 (C) 6 (D) 4

Ans. D

Sol. $(\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10$

$$\left((\sqrt{3} + \sqrt{2})^2 \right)^{\frac{x^2-4}{2}} + \left((\sqrt{3} - \sqrt{2})^2 \right)^{\frac{x^2-4}{2}} = 10$$

$$\Rightarrow (5 + 2\sqrt{6})^{\frac{x^2-4}{2}} + (5 - 2\sqrt{6})^{\frac{x^2-4}{2}} = 10$$

$$\text{Now } (5 + 2\sqrt{6})(5 - 2\sqrt{6}) = 25 - 24 = 1$$

$$\& (5 + 2\sqrt{6}) + (5 - 2\sqrt{6}) = 10$$

$$\therefore \frac{x^2-4}{2} = \pm 1 \quad \Rightarrow x^2 = 4 \pm 2$$

$$\Rightarrow x = \pm\sqrt{6}, \pm\sqrt{2}$$

$$\therefore n(S) = 4$$

20. Let $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$, where the function f satisfies $f(x+y) = f(x)f(y)$ for all natural numbers x, y and

$f(1) = 2$. Then the natural number 'a' is :

माना $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$ है, जहाँ सभी प्राकृत संख्याओं x, y के लिए, फलन f , $f(x+y) = f(x)f(y)$ को संतुष्ट करता है

तथा $f(1) = 2$ है, तो प्राकृत संख्या 'a' बराबर है :

- (A) 2 (B) 3 (C) 16 (D) 4

Ans. B

sol. $\therefore f(x+y) = f(x) \cdot f(y)$

$\therefore$ Let $\boxed{f(x) = b^x}$

$\therefore f(1) = 2$

$\therefore b = 2$

$\Rightarrow \boxed{f(x) = 2^x}$

Now, $\sum_{k=1}^{10} 2^{a+k} = 16(2^{10} - 1)$

$\Rightarrow 2^a \sum_{k=1}^{10} 2^k = 16(2^{10} - 1)$

$\Rightarrow 2^a \times \frac{(2^{10}) - 1}{2 - 1} \times 2 = 16 \times (2^{10} - 1)$

$\Rightarrow 2^a = 8$

$\Rightarrow \boxed{a = 3}$